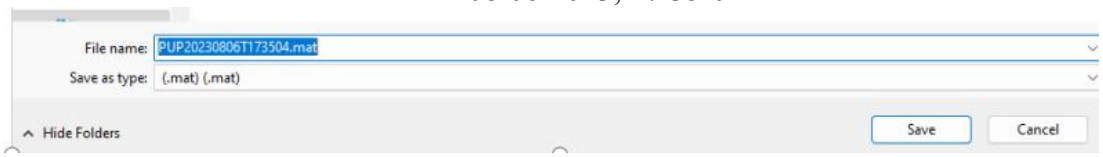
Data recording operation manual

1. Preparing

(1) Run Matlab GUI, push Refresh button to connect to radar device, push Start button to start.
(2) Set Transmitter, Receiver, modulation, Bandwidth, sweep Time, Sampling Number etc.

2. Recording

(1) Select Recording Time. for example: 10s
(2) Push Record button, the display window will stop displaying and start to record raw data.
(3) Waiting for 10s, a file save window will appear with default name like PUP20230806T173504.mat, Which includes the date and time information: 08/06/2023, 17:35:04.



(4) You can use the default name or assign a new name to it and push save button.
(5) The GUI will restore its display after file is saved.

3. Data loading

(1) Use "load" command in Matlab command window to read the raw data in. For example:
"Load PUP20230806T173504"

After this load command, the raw data matrix and some necessary variables are loaded automatically.
(2) Use "whos" to observe the data and variables loaded.
When Transmitter select Tx1, the list is:

```
>> whos
  Name                 Size                 Bytes  Class     Attributes

  ComplexDataTx1      256x15946            65314816  double    complex
  DATE                  1x20                   40  char
  LAFH                  1x1                     8  double
  LAFL                  1x1                     8  double
  LAMS                  1x2                     4  char
  LANR                  1x1                     8  double
  LANT                  1x1                     8  double
  LARS                  1x3                     6  char
  LASN                  1x1                     8  double
  LAST                  1x1                     8  double
  LATS                  1x3                     6  char
```

When Transmitter select Tx2, the list is:

```
>> whos
  Name                 Size                 Bytes  Class     Attributes

  ComplexDataTx2      256x15946            65314816  double    complex
  DATE                  1x20                   40  char
  LAFH                  1x1                     8  double
  LAFL                  1x1                     8  double
  LAMS                  1x2                     4  char
  LANR                  1x1                     8  double
  LANT                  1x1                     8  double
  LARS                  1x3                     6  char
  LASN                  1x1                     8  double
  LAST                  1x1                     8  double
  LATS                  1x3                     6  char
```

When Transmitter select Tx1.Tx2, the list is:

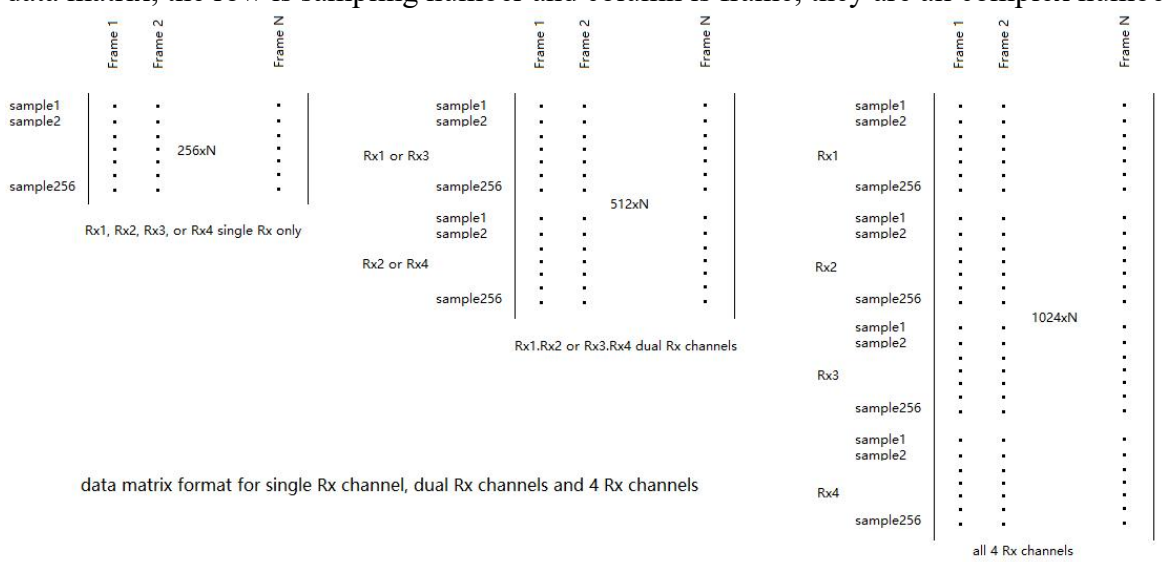
```
>> whos
  Name                Size                Bytes  Class     Attributes

  ComplexDataTx1      256x17992           73695232  double    complex
  ComplexDataTx2      256x17992           73695232  double    complex
  DATE                1x20                      40  char
  LAFH                1x1                        8  double
  LAFL                1x1                        8  double
  LAMS                1x2                        4  char
  LANR                1x1                        8  double
  LANT                1x1                        8  double
  LARS                1x3                        6  char
  LASN                1x1                        8  double
  LAST                1x1                        8  double
  LATS                1x7                       14  char
```

The definition of variables are:

DATE:	data file saving time, example:	DATE =”06-Aug-2023 13:29:54”
LAFH:	upper frequency, example:	LAFH = 2.5000e+10
LAFL:	lower frequency, example:	LAFL = 2.4000e+10
LAMS:	modulation, example:	LAMS=”sawtooth”
LANR:	number of receivers, example:	LANR =1
LANT:	number of transmitters, example:	LANT =1
LARS:	Receiver Name String, example:	LARS =”Rx1”
LASN:	Sampling number, example:	LASN =256
LAST:	Sweep Time, example:	LAST =5.000E-04
LATS:	Transmitter Name String, example:	LATS =”Tx1”

For the data matrix, the row is sampling number and column is frame, they are all complex number.



data matrix format for single Rx channel, dual Rx channels and 4 Rx channels

4. Data extraction code examples (ExtractRecordData.m):

```
load PUP20230806T192612
Tx1Rx1=ComplexDataTx1(1:256, :);
Tx1Rx2=ComplexDataTx1(257:512, :);
figure(1)
plot(abs(Tx1Rx1(:, 16)))
figure(2)
plot(abs(Tx1Rx2(:, 200)))
```